QP CODE: 24900180

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MC,U-UGP (HONOURS) REGULAR
EXAMINATION NOVEMBER 2024
First Semester
Multi-Disciplinary Course - MGI MDCMBG,100 - FASCINATING WORLD OF MICROBES
(2024 ADMISSION ONWARDS)

Maximum Marks: 35 **Duration: 1.5 Hours**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Answer any l0 out of 12
Each question carries l marks

1. **Who performed swan neck experiment to disprove spontaneous generation theory?** **[K] [1]**
   Louis Pasteur (0.5). Give a note on spontaneous generation theory.(0.5)
2. **What are Animalcules**? **[K] [2]**
   Term Animalcules introduced by Antony van Leeuwenhoek(0.5). Define animalcules(0.5)
3. **Name the scientist who invented microscope?** **[K] [1]**
   Antony van Leeuwenhoek(0.5).) Define animalcules(0.5)
4. **Name two vaccines developed by Louis Pasteur?** **[K] [1]**
   Cholera vaccine (0.5), rabies vaccine(0.5)
5. **Who discovered penicillin?** **[K] [1]**
   Alexander Fleming(0.5). Write a note on discovery.(0.5)
6. **List two examples for Bacteria?** **[K] [1]**
   Name any two bacteria.(0.5+0.5)
7. **Name the organic acid content of vinegar.** **[K] [2]**
   Acetic acid (0.5)Write a note on vinegar (0.5)
8. **What is brine?** **[K] [2]**
   Define brine(0.5). Use (0.5)
9. **What is Penicillin?** **[K] [2]**
   Antibiotic(0.5), Write a note on discovery(0.5)
10. **Outline the importance of Nostoc.** **[U] [2]**
    Nitrogen fixation (0.5), Note on Nostoc)(0.5)
11. **Name the common disease caused by E.coli.** **[K] [2]**
    Name atleast 2 diseases(0.5+0.5). UTI,Diarrhea, Septcemia, Meningitis,Pyogenic infections
12. **List out two harmful fungus.** **[A] [2]**
    Name any two harmful fungus(0.5+0.5)

Part B
Answer any 3 out of 6
Each question carries 5 marks

13. **What is Abiogenesis? Name the scientists who supported and opposed** **[K] [1]**

spontaneous generation

Define Abiogenesis(1), Name any two scientists who supported spontaneous generation with their experiment (2) and Name any two scientists who opposed spontaneous generation with their experiment (2)

14. State the characteristics of Kingdom Monera. [K] [1]

Kingdom including prokaryotes.(1).Write at least 5 characteristics of Kingdom Monera (4)

15. Differentiate between symbiotic and nonsymbiotic nitrogen fixers. [A] [2]

Define symbiotic nitrogen fixers with example (1), Define nonsymbiotic nitrogen fixers with example (1). Write at least 3 differences between symbiotic and nonsymbiotic nitrogen fixers.

Nitrogen-fixing bacteria coexist symbiotically with the host plant whereas nonsymbiotic nitrogen fixers free-living bacteria in the soil.

Symbiotic -Nodule formation , Nonsymbiotic – No Nodule formation

Symbiotic nitrogen fixers – Seen in Rhizosphere, Nonsymbiotic nitrogen fixers everywhere

16. Explain the method of Isolation and identification of Rhizobium? [U] [2]

Define Rhizobium (0.5) .Steps in isolation of rhizobium from root nodule (3) identification (1.5)

17. Explain Biopesticides with any two examples? [U] [2]

Define Biopesticide(1) Advantages (2.5) Give 2 Examples(1.5)

18. Explain source, transmission and control of human corona virus? [E] [2]

Source(1), transmission and symptoms (2) and control – prophylaxis and vaccination (2)

Part C

Answer any 1 out of 2

Each question carries I0 marks

19. Describe Whittaker's five kingdom classification with reference to their characteristic features. Give examples. [K] [1]

Introduced by R.H Whittaker (0.5), Demerits of 3 domain classification which led to Five kingdom classification (2). Explain each kingdom with example (1.5*5=7.5)

20. Explain fermented foods, types and advantages. [E] [2]

Define fermented foods.(1) Advantages of fermented foods (2), Write in detail the production and importance of any 2 fermented foods.(3.5*2=7)